Mathematics - Undergraduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Determine whether the polynomial in Z[x] satisfies an Eisenstein criterion for irreducibility over Q. x^2 - 12
   (a) Yes, with p=2.
   (b) Yes, with p=3.
   (c) Yes, with p=5.
   (d) No.

2. Statement 1 | Every nonzero free abelian group has an infinite number of bases. Statement 2 | Every free abelian group of rank at least 2 has an infinite number of bases.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

3. Suppose A, B, and C are statements such that C is true if exactly one of A and B is true. If C is false, which of the following statements must be true?
   (a) If A is true, then B is false.
   (b) If A is false, then B is false.
   (c) If A is false, then B is true.
   (d) Both A and B are true.

4. Let f(x) be a function whose graph passes through the origin. If f(2 n) = n^2 + f[2(n - 1)] for every integer n, what is the value of f(8)?
   (a) 24
   (b) 30
   (c) 32
   (d) 36

5. Find the characteristic of the ring Z x Z.
   (a) 0
   (b) 3
   (c) 12
   (d) 30

True / False (10 marks)

Answer True or False

6. True or False: The sum of the determinants of matrix A and matrix B is equal to -1.
7. True or False: The polynomial f(x) = x^3 + 5 x^2 + 4 x + 1 can have complex roots 2 + i and 1 - i, as it is of sufficient degree.
8. True or False: The solution of the differential equation gives f(2) as 1/e.
9. True or False: The polynomial x^3 + 2 x^2 + 2 x + 1 can be factored into linear factors as (x - 1)(x - 4)(x - 2) in Z_7[x].
10. True or False: If f(z) is an analytic function mapping the complex plane to the real axis, the imaginary axis is mapped to the entire real axis.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Abigail, Beatrice, and Carson combine their eggs to sell them at the market. If Abigail has 37 eggs, Beatrice has 49 eggs, and Carson has 14 eggs, and if eggs can only be sold in cartons of 12, how many eggs will be left over if all cartons are sold?
12. Given a finite set X, where the number of subsets containing exactly 3 elements exceeds those containing exactly 2 elements by 14, determine how many subsets of X contain exactly 4 elements. Justify your reasoning.

End of Paper